

Partial differential Eqn

Consider a general Partial diff eqn of second order for a f'n of two independent, variables x & y in the form

$$Rr + Ss + Tt + f(x, y, z, t, q) = 0$$

where R, S, T are continuous f'n of x & y . and

$$r = \frac{\partial^2 u}{\partial x^2}, \quad s = \frac{\partial^2 u}{\partial x \partial y}, \quad t = \frac{\partial^2 u}{\partial y^2}$$

then the PDE is said to be

1) Hyperbolic at a point (x, y) in D if $S^2 - 4RT > 0$.

2) Parabolic at a point (x, y) in D if $S^2 - 4RT = 0$.

3) Elliptic at a point (x, y) in D if $S^2 - 4RT < 0$.

$$\underline{Q} \quad (1) \quad 2 \frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial^2 v}{\partial x \partial y} + 3 \frac{\partial^2 w}{\partial y^2} = 2$$

Sol

$$2R + 4S + 3T = 2$$

Here $R=2$, $S=4$, $T=3$ (on comparing $Rx^2 + Sxy + Ty^2 + \dots$)

$$\begin{aligned} S^2 - 4RT &= 16 - 4 \times 2 \times 3 \\ &= 16 - 24 \\ &= -8 < 0 \end{aligned}$$

hence Elliptic.

$$(2) \quad x - t = 0$$

Sol $R=1$, $S=0$, $T=-1$

$$S^2 - 4RT = 0 - 4 \times 1 \times (-1) = 4 > 0$$

hyperbolic.

$$3) \quad \frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial^2 v}{\partial x \partial y} + 4 \frac{\partial^2 w}{\partial y^2} = 0$$

Sol $R=1$, $S=4$, $T=4$

$$\begin{aligned} S^2 - 4RT &= 16 - 4 \times 1 \times 4 \\ &= 16 - 16 \\ &= 0 \end{aligned}$$

$$\therefore S^2 - 4RT = 0$$

hence Parabolic.

$$4) \quad xyx - (x^2 - y^2)s - xyt + by - qx = 2(x^2 - y^2)$$

Sol Here $R = xy$, $S = -(x^2 - y^2)$, $T = -xy$

$$S^2 - 4RT = [-(x^2 - y^2)]^2 + 4 \cdot xy \cdot xy$$

$$= (x^2 - y^2)^2 + 4x^2y^2$$

$$= x^4 - 2x^2y^2 + y^4 + 4x^2y^2$$

$$= x^4 + 2x^2y^2 + y^4$$

$$= (x^2 + y^2)^2 > 0$$

So the given pde is hyperbolic.

$$5) \quad x^2(y-1)x - x(y^2-1)s + y(y-1)t + xyb - q = 0$$

classify the following eqns: -

$$1) \quad 2 \frac{\partial^2 y}{\partial x^2} + 6 \frac{\partial^2 y}{\partial x \partial y} + 3 \frac{\partial^2 y}{\partial y^2} = 0$$

$$2) \quad \frac{\partial^2 y}{\partial x^2} + 4 \frac{\partial^2 y}{\partial x \partial y} + 4 \frac{\partial^2 y}{\partial y^2} = 0$$

1) Here $R=2$, $S=6$, $T=3$

$$S^2 - 4RT = 36 - 4 \times 2 \times 3 \\ = 36 - 24 = 12 > 0$$

hence hyperbolic.

Examine which of the following are hyperbolic, parabolic and elliptic?

a) $x^2 \frac{\partial^2 y}{\partial t^2} - \frac{\partial^2 y}{\partial n^2} = u$

b) $t \cdot \frac{\partial^2 y}{\partial t^2} + 2 \frac{\partial^2 y}{\partial n \partial t} + x \frac{\partial^2 y}{\partial n^2} + \frac{\partial y}{\partial n} = 0$

b) sol Here $R=x$, $S=2$, $T=t$

$$S^2 - 4RT = 4 - 4xx \times t \\ = 4 - 4xt$$

it is hyperbolic if $S^2 - 4RT > 0$

i.e. $4 - 4xt > 0$ i.e. $tx < 1$.

elliptic if $4 - 4xt < 0$ i.e. $tx > 1$

and parabolic if $4 - 4xt = 0$ i.e. $tx = 1$

classify each of the following:-

$$1) \quad 9 \frac{\partial^2 y}{\partial x^2} - 6 \frac{\partial^2 y}{\partial x \partial t} + \frac{\partial^2 y}{\partial t^2} = 0$$

$$2) \quad 3 \frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + 4 \frac{\partial^2 z}{\partial y^2} = 0$$

$$3) \quad 2 \frac{\partial^2 z}{\partial x^2} - 6 \frac{\partial^2 z}{\partial x \partial y} + 3 \frac{\partial^2 z}{\partial y^2} = 0$$

$$4) \quad \frac{\partial^2 z}{\partial x^2} = \frac{\partial x}{\partial y}$$

$$5) \quad \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$$

$$6) \quad \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$